

increase in employment in software development occupations since the release of ChatGPT.

There are a variety of occupation codes that relate to computer programming, corresponding to codes 1005 through 1108. While software developers represent the largest share of employment in this group, one possibility is that employment gains for software developers come from reclassification of workers away from other coding occupations to software development.⁷ Figure 2 shows the employment and earnings trends for all computer occupations aggregated together. The results again suggest little evidence of a decline in employment or real earnings.

The other series in Figure 2 show employment and earnings trends for customer service representatives, another occupation in the top quartile of AI exposure. Brynjolfsson et al. (2025) find that access to AI assistance increases worker productivity, as measured by issues resolved per hour, by 15% on average for customer service representatives, with substantial heterogeneity across workers. Figure 2 suggests that employment in customer service representatives has declined since the release of ChatGPT.

These findings raise the question of which occupational characteristics may influence the employment effects of AI given some level of measured exposure. Figure 3 shows employment trends for the most exposed quartile of occupations split by whether the college share in the occupation is above or below the median. The results suggest employment share increases in more educated exposed occupations and employment share declines in less educated exposed occupations in the periods before and after the release of ChatGPT.

4.3 Labor Market Trends for Entry-Level Workers

In recent months there has been a growing concern that AI is beginning to displace entry-level workers (Thompson, 2025; Raman, 2025; Allen, 2025). The top panel of Figure 4 shows the employment share trends for entry level versus experienced workers. The analysis shows that entry level workers have cyclical and volatile employment compared to more experienced

⁷See Van Dam (2025) for a discussion of trends across different computer occupations.

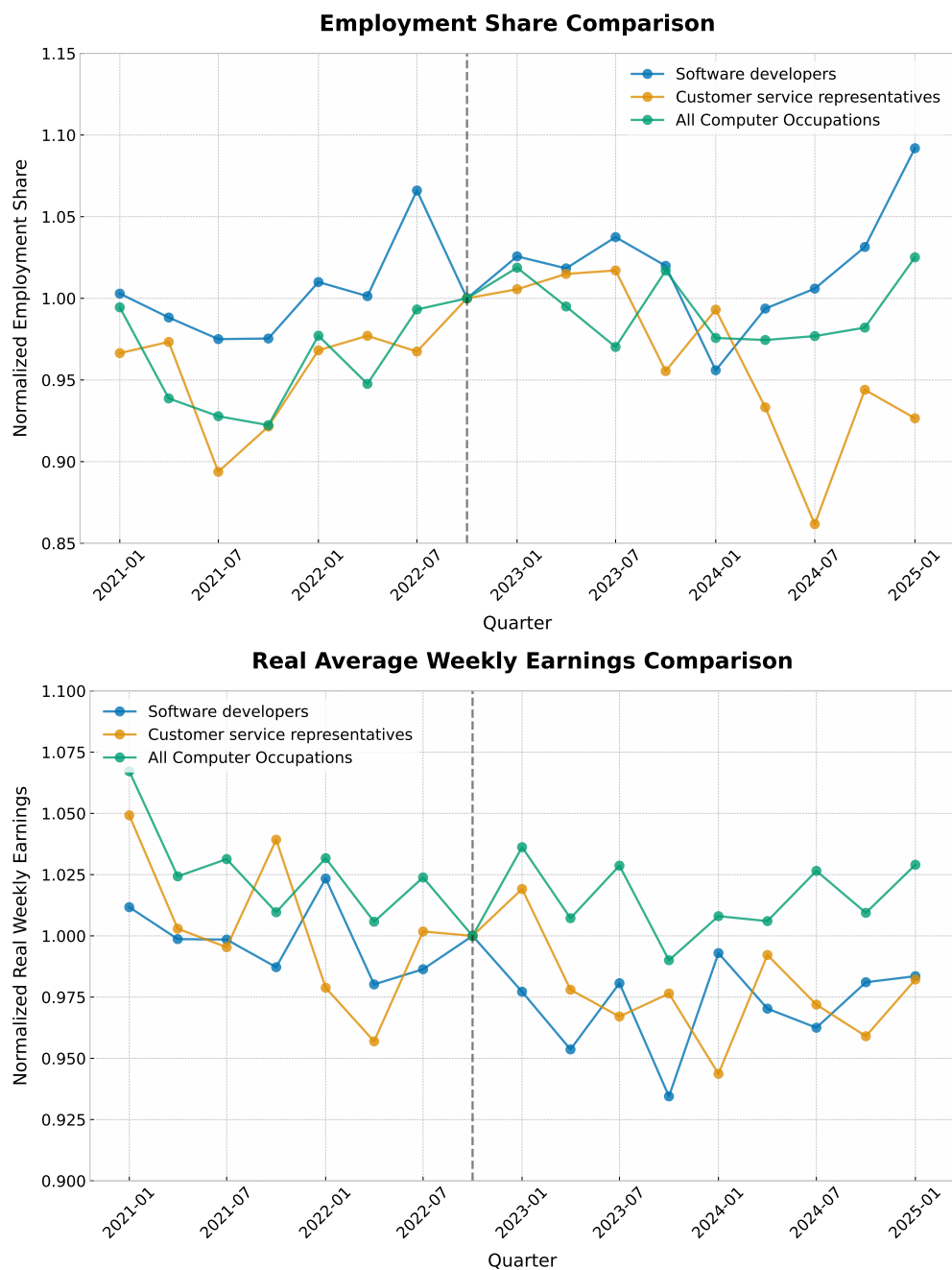


Figure 2: Employment share and real earnings trends for software developers, computer occupations, and customer service representatives. Computer occupations include all occupations with Census occupation codes 1005 through 1108.

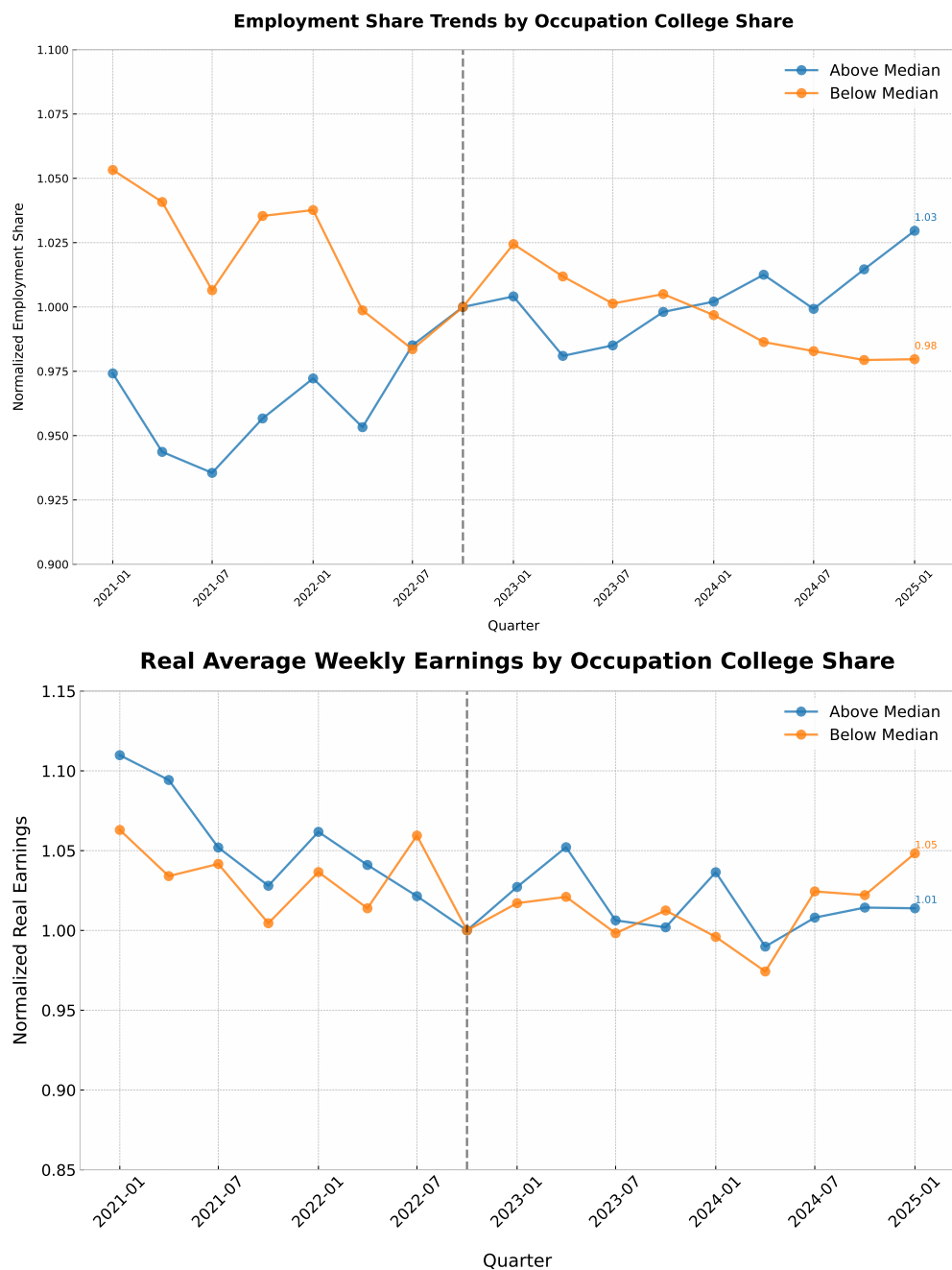


Figure 3: Employment share and real earnings trends for the most exposed quartile of occupations, split by whether the occupation has above median or below median college share.

workers, with no notable overall trend in employment shares over the sample period. The bottom panel likewise shows no evident trend in real earnings.

These patterns for entry level workers overall might mask trends in occupations more exposed to AI. Figure A4 shows the employment share trends for entry level versus experienced workers in the quartile of occupations most exposed to AI according to the GPT-4-based β measure from Eloundou et al. (2024). Figure A5 similarly shows the employment trends for entry level versus experienced workers in computer occupations. Both plots are noisy, primarily because restricting to workers under the age of 26 reduces the sample size. With more time and larger surveys it may become apparent whether trends are sustained.

5 Relationship between Job Postings and Employment

The general purpose nature of artificial intelligence has raised concerns of widespread employment declines, with some speculation that these trends are already visible within certain industries such as software development. One piece of evidence taken to suggest employment declines already are in job posting trends from sources such as Indeed. As shown in Figure 5, the number of job postings for software developers has fallen since a peak in early 2022.

Three aspects of the data suggest caution in interpreting the job posting trends to signal labor market disruption for software developers caused by AI.

1. The number of job postings has fallen for a range of other occupations as well, including in professions such as nursing with low estimated AI exposure. This suggests that the decline is not specific to software developers.
2. While the decline is steeper for software developers than other professions, the peak in early 2022 is also considerably higher. This could indicate macroeconomic trends in the tech sector that led to a boom and bust in job postings unrelated to AI.
3. The decline in job postings is not matched by a decline in employment in the software development occupation in the CPS, as shown in Figure 2. If anything there is some